BAHRIA UNIVERSITY ISLAMABAD

Department of Computer Science • Mobile Application Development (MAD)

Assignment 2 — UI & Front-End Submission

KharchaBook — Student Expense Manager

Complete Layout Design, Navigation Flow, Styling System & UI Code (Zero Backend)

Course & Section:	Mobile Application Development (MAD) — BS-IT 6A
Group Members:	1. Zain Kashif (Enrollment: 01-135232-107) 2. Huraira Minhas (Enrollment: 01-135232-030)
Approved Project Type:	Single-User Student Expense Tracker (Flutter Client + Supabase Backend)
Deliverable Type:	Assignment 2: Complete User Interface (UI Code Only, No Backend)
Submission Date:	November 2026 (Semester Fall 2026 — Submitted via LMS)

1. Executive Summary & Assignment 2 Scope

Assignment 2 Purpose: This submission satisfies all requirements of Assignment 2 (UI Submission) for Mobile Application Development (MAD) at Bahria University Islamabad. As specified by the department and instructor, this submission focuses exclusively on the complete, fully realized front-end User Interface (UI) of our approved project proposal, KharchaBook — Student Expense Manager, without requiring any backend or database functionality at this stage.

App Core Concept: KharchaBook is a dedicated expense tracking solution tailor-made for university students (specifically undergraduates living on fixed pocket-money allowances or hostel mess stipends). In this milestone, we deliver all 8 screens proposed in Assignment 1 with pixel-perfect visual layouts, fluid inter-screen navigation, custom vector illustrations, tactile data visualizations, responsive adaptability across mobile screen form factors, and complete UI-only Dart source code.

1.1 Approved Proposal Mapping & Deliverable Matrix

The table below details how every screen and feature proposed in Assignment 1 has been designed and implemented for Assignment 2:

ID	Screen Name	UI Layout & Key Elements	Proposal Reference
S1	Splash Screen	Illustrated student desk scene, monogram badge, title, semester badge, 2s session router	Clause 7 (S1 Splash)
S2	Student Login	KB logo pill, student email field, obscure password, error banner, register redirection	Clause 7 (S2 Login)
S3	Student Register	Full name, university email, password + confirm, validation rules, sign-in link	Clause 7 (S3 Register)
S4	Main Dashboard	Header with avatar, search pill, category tags, Hero Allowance card, Calendar strip, grouped feed	Clause 7 (S4 Dashboard)
S5	Add/Edit Expense	Numeric PKR hero input, quick increments (+50..1000), category selector, date picker, delete modal	Clause 7 (S5 Add/Edit)
S6	Visual Analytics	Hero card, period selector, Daily Spending Trend chart, tactile Dot Matrix Breakdown	Clause 7 (S6 Stats)
S7	Profile & Settings	Student pass card, campus badge, live theme switcher (OLED Dark vs Pastel), budget dialog	Clause 7 (S7 Profile)
S8	Transaction Ledger	Dedicated history list with	Clause 6 (F3 Filter/Ledger)

search bar, type filter chips (All/Expense/Income), totals strip

2. Visual Architecture & Design System

Dual-Theme Architecture: To elevate KharchaBook beyond conventional AI-generated or generic mobile templates, we created a coherent, bespoke Design System inspired by modern editorial fintech publications and student-centric productivity tools. The app features a complete Dual-Theme architecture allowing users to switch seamlessly between two calibrated palettes directly from the Profile screen:

- **1. Editorial Pastel Mode (Light Default):** Constructed with a calming warm paper canvas (#F7F6F2), deep ink-black typography (#111827), and purposeful pastel accents: Buttercup Yellow (#FDE047) for semester allowance, Lilac Lavender (#DDD6FE) for academic discovery, Soft Mint (#A7F3D0) for income/safety, and Soft Coral (#FECDD3) for expense warnings. Accompanying vector illustrations feature 2.2px solid ink outlines and zero clipping.
- **2. OLED Dark Neo-Fintech Mode (Dark):** Built for battery efficiency and high-contrast night study sessions. Utilizes a true OLED canvas (#0A0B0E), dark elevated card surfaces (#151821), signature electric Volt Lime accents (#D4F938), soft neon glow filters, and sharp contrast text.

2.1 Complete Color Palette Specification

The color architecture is declared centrally in core/theme/app_colors.dart with semantic roles across both themes:

Token Name	Hex Code	Light Role	Dark Role	Semantic Purpose
Canvas / Background	#F7F6F2 / #0A0B0E	Warm Off-White	True OLED Pitch Black	App surface backdrop
Card Surface	#FFFFFF / #151821	Pure White + Shadow	Elevated Dark Slate	Content card enclosures
Primary Text	#111827 / #F8FAFC	Solid Ink Black	Crisp Clean White	Headings, titles, balances
Secondary Text	#64748B / #94A3B8	Slate Muted Gray	Slate Medium Gray	Subtitles, captions, dates
Accent Volt Lime	#D4F938	Secondary Indicator	Primary Glow / Active Tab	Active badges & primary buttons
Pastel Yellow	#FDE047	Budget Hero Card	N/A (Editorial Light)	Semester allowance container
Pastel Lavender	#DDD6FE	Analytics Hero Card	N/A (Editorial Light)	Breakthrough discovery card
Income Green / Expense Red	#10B981 / #F43F5E	Emerald / Vivid Rose	Neon Green / Coral Rose	Financial inflow vs outflow

2.2 Bespoke Interactive UI Components

- **Arched Floating Navigation Dock:** A custom-drawn floating navigation bar using CustomPainter (ArchedDockPainter). Features continuous cubic Bezier curves creating a subtle dome peak in the horizontal center, flanked by 4 symmetric navigation tabs (Home, Analytics, History, Profile) and an elevated 52x52 action button (+) that triggers the transaction modal.

- **Tactile Circular Dot Matrix Chart:** A custom visual data presentation inspired by tactile analog meters. Renders a circular ring of 44 discrete circular dots colored proportionally by category spending. In light mode, unused dots rest in soft slate (#E2E8F0); in dark mode, active dots radiate vivid neon highlights.
- **Dynamic Daily Spending Trend Chart:** An interactive dual-view bar chart (Daily vs Weekly view toggle) tracking daily burn rate against the daily allowance target limit. Bars within safe pace render in editorial purple/mint, while spending spikes crossing the daily threshold automatically highlight in warning rose with dashed budget threshold lines.
- **Weekly Calendar Strip:** Horizontal interactive day-picker widget allowing students to jump to any day of the current week to inspect daily cash expenses.
- **Unclipped Student Vector Illustrations:** Hand-crafted CustomPainters rendering student-themed scenes: graduation caps, semester notebooks with bookmarks, campus chai mugs, student ID cards, and study desks. Uses fixed Row proportions so graphics never clip across devices.

3. Screen Navigation Architecture & Routing Map

Navigation Structure: The navigation architecture implements a unidirectional and hierarchical state flow. The application root initializes with a MultiProvider maintaining ThemeProvider, AuthProvider, and ExpenseProvider.

1. Splash (S1) is the application entrypoint. It runs a 2-second branded splash animation and evaluates session state:
   • If unauthenticated -> redirects to LoginScreen (S2).
   • If authenticated -> redirects directly to MainShellScreen (S4–S8).
2. Login (S2) and Register (S3) transition bi-directionally with smooth slide animations.
3. MainShellScreen maintains an IndexedStack preserving screen state across the 4 primary tabs:
   • Tab 0: DashboardScreen (S4)
   • Tab 1: StatsScreen (S6)
   • Tab 2: TransactionsScreen (S8)
   • Tab 3: ProfileScreen (S7)
4. Centered '+' Floating Button pushes AddEditExpenseScreen (S5) as a full-height animated modal sheet from any tab.
5. Tapping any existing transaction item navigates into AddEditExpenseScreen in edit mode, pre-filling its fields and presenting a delete option.
6. SearchFilterBottomSheet is presented as a modal bottom sheet from either the Dashboard or Ledger screen.

4. Screen-by-Screen UI Showcase & Visual Evidence

Below is the comprehensive visual evidence of all screens implemented in KharchaBook, with high-resolution device captures taken from the release APK on Android:



Figure 1: S1 — Illustrated Student Splash Screen

Screen S1: Illustrated Student Splash Screen

The entrypoint screen sets an authentic university student tone.

Key Visual Elements:
• Custom editorial illustration of a student at a study desk with semester textbooks, university chai mug, laptop, and starburst sparkles.
• Embossed 'KB' monogram brand badge.
• High-contrast bold typography: 'KharchaBook — Student Expense Manager'.
• Semester section badge: 'Bahria University Islamabad • BS-IT 6A'.
• Smooth entrance fade transition and 2-second session redirection timer.

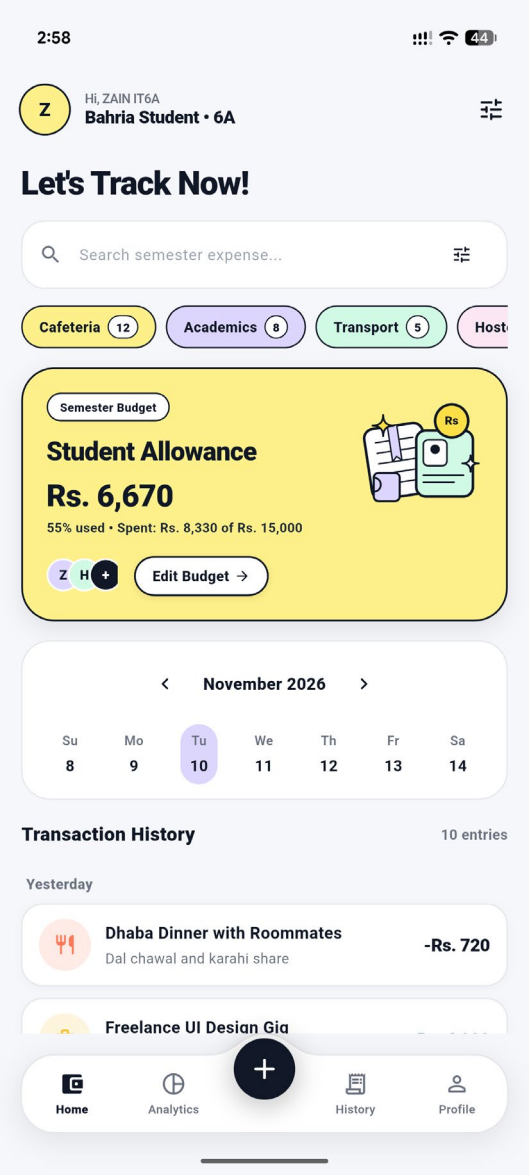


Figure 2: S4 — Student Expense Dashboard (Editorial Pastel)

Screen S4: Dashboard — Editorial Pastel Light Mode

The primary student home hub providing instant budget awareness under 10 seconds.

Key Visual Elements:
• Personalized greeting: 'Hi, ZAIN IT6A • Bahria Student 6A' with circular avatar.
• Bold headline: 'Let's Track Now!' with search bar pill.
• Quick category filter pills with transaction counts: Cafeteria (12), Academics (8), Transport (5), Hostel (4).
• Buttercup Yellow Hero Card displaying remaining semester budget (Rs. 6,670), percentage consumed (55%), unclipped student ID & ledger illustration, and 'Edit Budget' pill.
• Weekly Calendar Strip for date filtering.
• Arched floating navigation dock at bottom with elevated '+' button.

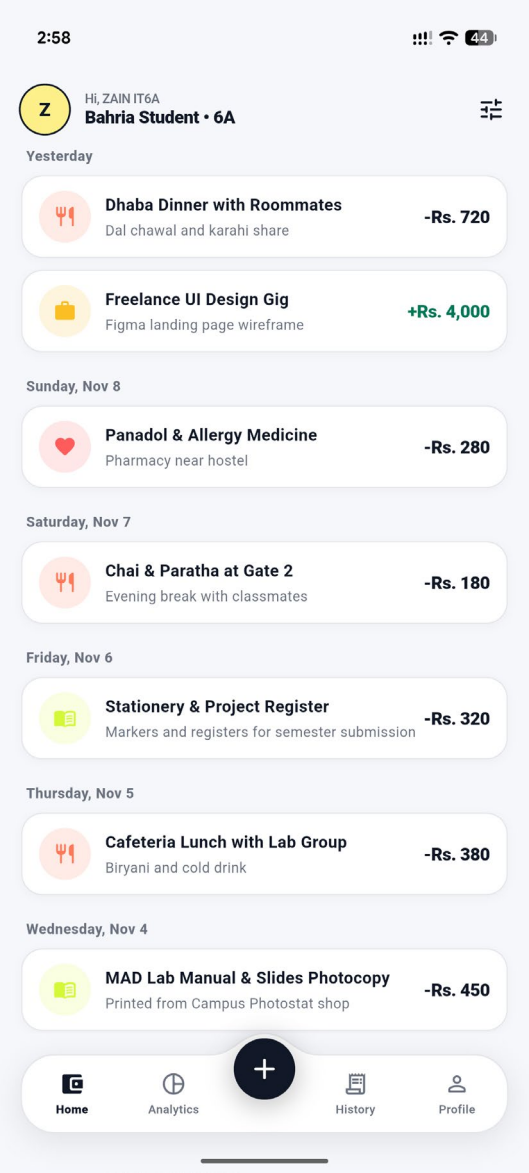


Figure 3: S4/S8 — Grouped Transaction History Feed

Screen S4 / S8: Grouped Transaction Feed

Chronologically grouped transaction ledger presenting daily cash outflows and inflows.

Key Visual Elements:

• Clean relative date headers ('Yesterday', 'Sunday, Nov 8', 'Saturday, Nov 7', etc.).

• Theme-aware white cards with subtle border outlines and soft drop shadows.

• Category-tinted avatar badges: Fork/Spoon (Food), Briefcase (Income), Heart (Health), Book (Academics).

• Clean subtitle notes: 'Dal chawal and karahi share', 'Pharmacy near hostel', 'Chai & Paratha at Gate 2', 'MAD Lab Manual Photocopy'.

• Color-coded amounts: Inflow in Emerald (+Rs. 4,000) and Outflow in Ink Black (-Rs. 720).

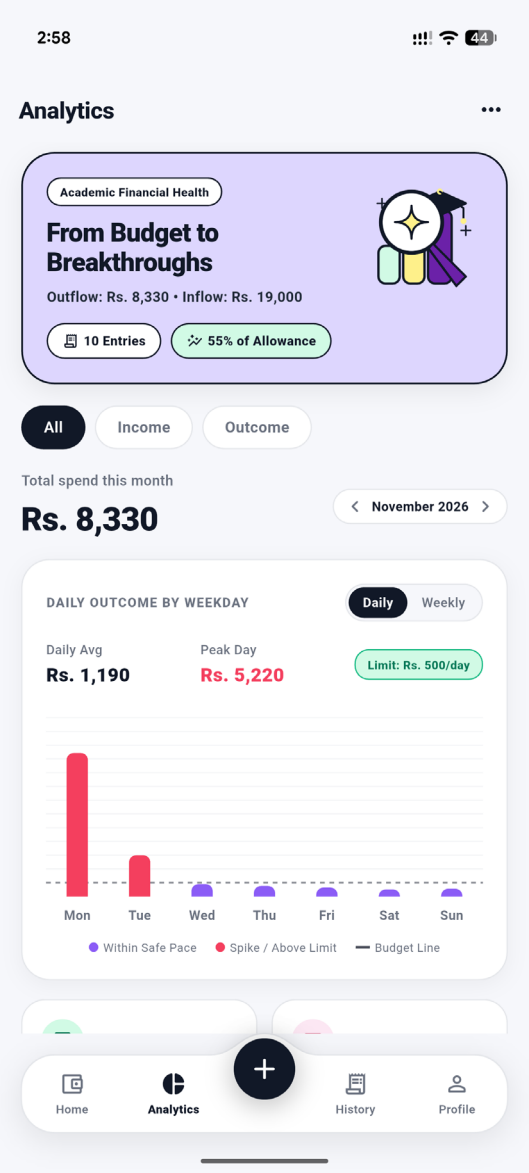


Figure 4: S6 — Visual Analytics Screen (Top Half: Trend Chart)

Screen S6: Analytics Screen — Daily Spending Trend

Comprehensive spending trend analytics providing early warnings before budgets are exceeded.

Key Visual Elements:
• Lavender Hero Card: 'Academic Financial Health — From Budget to Breakthroughs' with unclipped graduation cap and magnifying glass illustration.
• Summary badges: '10 Entries', '55% of Allowance'.
• Period switcher: 'All' | 'Income' | 'Outcome'.
• Total monthly spend display (Rs. 8,330) with month navigator.
• DAILY OUTCOME BY WEEKDAY Bar Chart: features Daily vs Weekly toggle, average calculation, peak day spike detection, Rs. 500/day dashed target budget line, and rose spike bars.

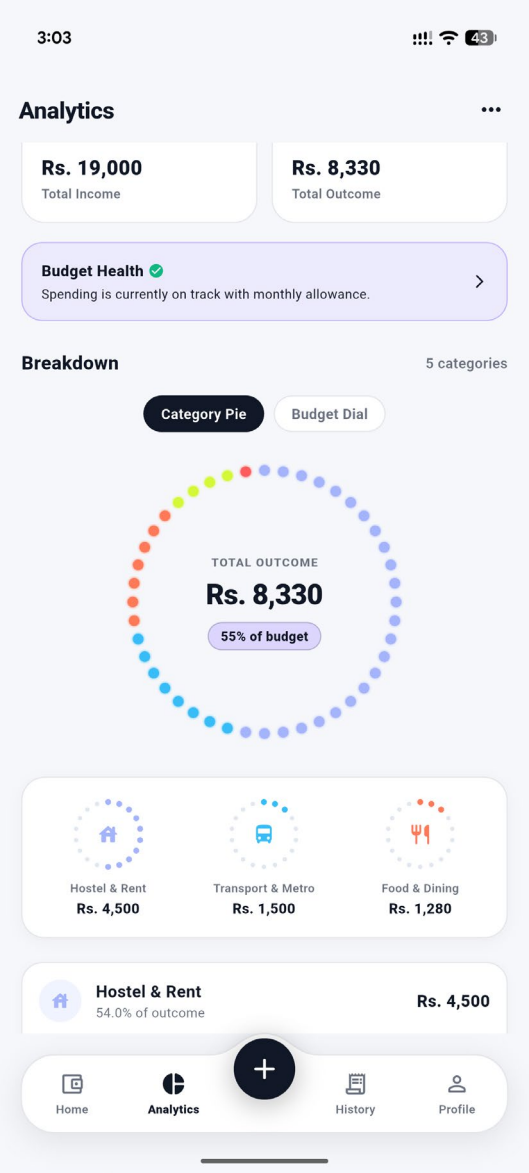


Figure 5: S6 — Visual Analytics Screen (Bottom Half: Breakdown)

Screen S6: Analytics Screen — Breakdown & Dot Matrix

Tactile category breakdown presenting financial distribution without clutter.

Key Visual Elements:

- Dual KPI summary cards: Total Income (Rs. 19,000) and Total Outcome (Rs. 8,330).
- Budget Health status banner with checkmark indicator.
- Renamed 'Breakdown' section header with category counter.
- Mode toggle pills: 'Category Pie' vs 'Budget Dial'.
- Tactile Circular Dot Matrix Pie Chart with discrete colored dots representing category proportions and center metric.
- Mini category dials: Hostel & Rent (Rs. 4,500), Transport (Rs. 1,500), Food & Dining (Rs. 1,280).

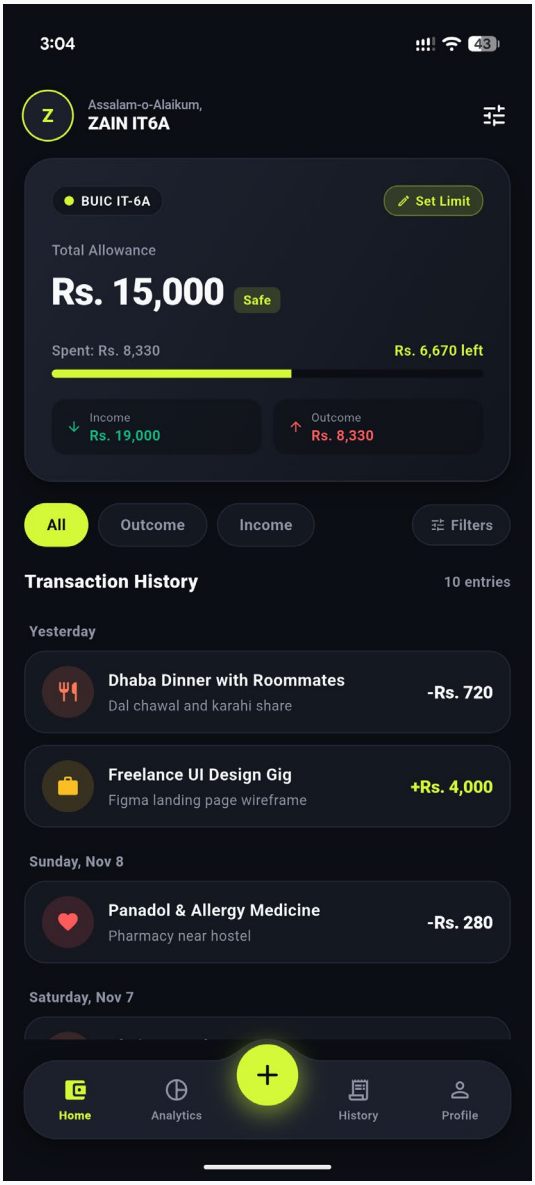


Figure 6: S4 — Student Expense Dashboard (OLED Dark Mode)

Screen S4: Dashboard — OLED Dark Neo-Fintech Mode

High-contrast dark mode for low-light environments and battery saving.

Key Visual Elements:
• Pitch-black canvas (#0A0B0E) with dark elevated cards (#151821).
• Monex-inspired virtual student pass card with BUIC IT-6A badge and electric Volt Lime allowance progress bar.
• Dual KPI pills displaying Rs. 19,000 income and Rs. 8,330 outcome.
• Pill filter row: 'All' (Volt Lime), 'Outcome', 'Income', 'Filters'.
• Dark transaction cards with muted borders.
• Arched bottom dock in dark slate with neon lime glowing elevated action button.

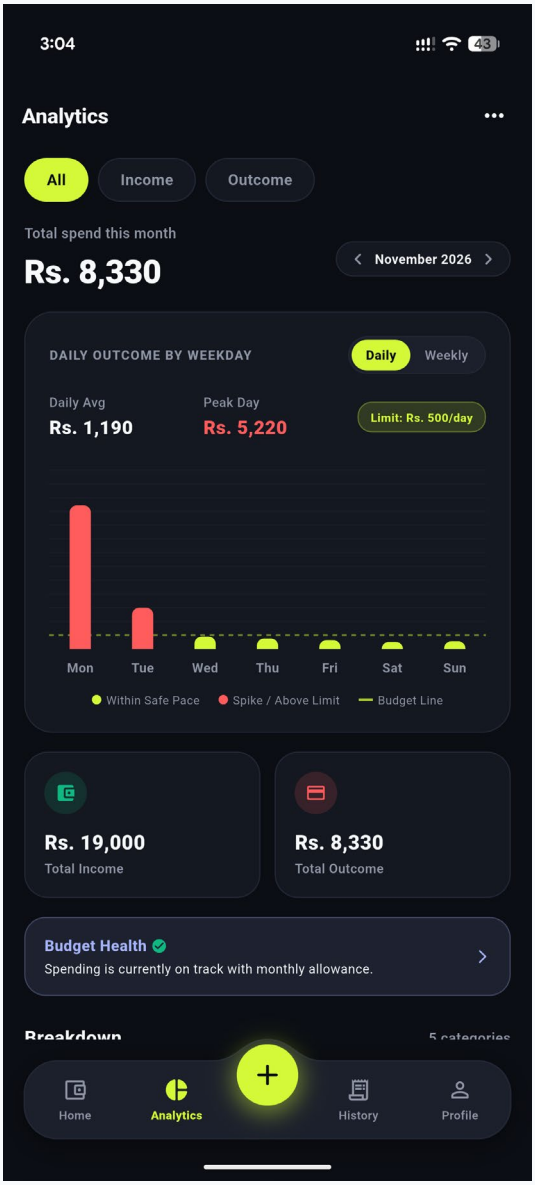


Figure 7: S6 — Visual Analytics Screen (OLED Dark Mode, Top Half)

Screen S6: Analytics — OLED Dark Trend Chart

Dark mode presentation of the dynamic weekday spending trend.

Key Visual Elements:

- High-contrast dark chart card with volt lime 'Daily' view selector.
- Real-time peak day identification (Rs. 5,220 on Monday) highlighted in coral red.
- Safe daily pace markers in electric volt lime on Wednesday through Sunday.
- Dashed lime budget threshold line set at Rs. 500/day allowance pace.
- Chart legend clearly explaining safe pace vs spike occurrences.

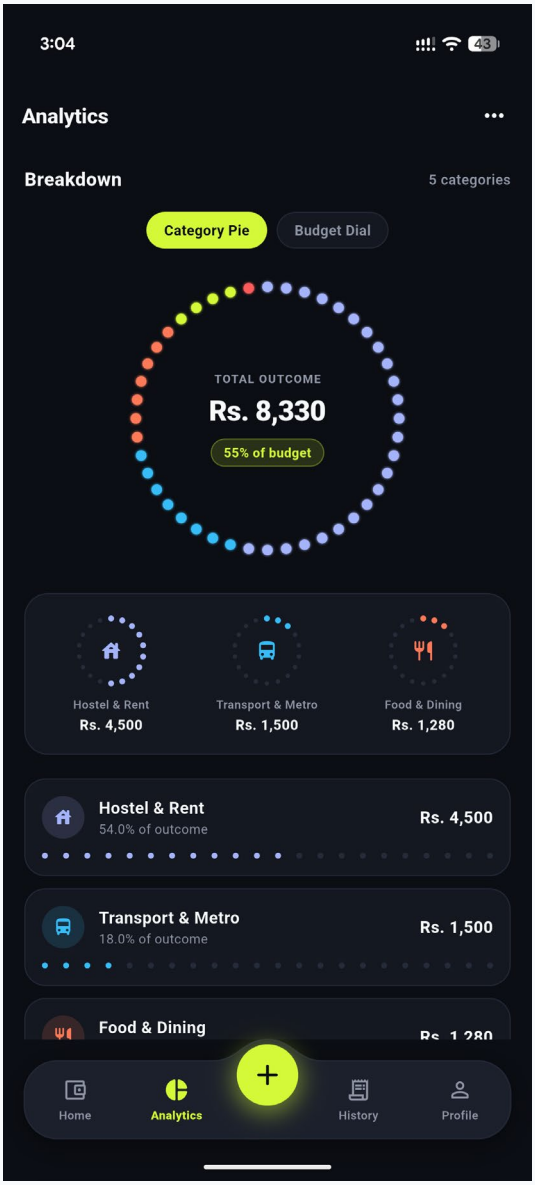


Figure 8: S6 — Visual Analytics Screen (OLED Dark Mode, Breakdown)

Screen S6: Analytics — OLED Dark Dot Matrix Breakdown

Tactile Dot Matrix Circular Ring Chart glowing against the deep OLED canvas.

Key Visual Elements:
• Renamed 'Breakdown' section header with 5 categories indicator.
• Glowing multi-colored discrete circular dots representing Hostel (blue), Food (coral/yellow), and Transport (cyan).
• Center metric display: TOTAL OUTCOME Rs. 8,330 • 55% of budget.
• Category breakdown cards with discrete 22-dot progress bars showing exact expenditure and percentage shares.

5. Complete UI Source Code (UI Only, No Backend)

UI Source Code Architecture: In strict accordance with Assignment 2 instructions, only front-end User Interface code is submitted below. The UI is completely decoupled from backend persistence, utilizing clean repository abstractions and local mock providers for state simulation.

5.1 Centralized Palette & Styling Tokens (app_colors.dart)

Declares all semantic color constants for both OLED Dark and Editorial Pastel themes:

Listing: lib/core/theme/app_colors.dart

```
import 'package:flutter/material.dart';

class AppColors {
  AppColors._();

  // Signature Volt Lime & OLED Dark (Neo-Fintech Theme)
  static const Color neonLime = Color(0xFFD4F938);
  static const Color neonLimeDark = Color(0xFFB5D826);
  static const Color neonLimeSoft = Color(0x26D4F938);

  static const Color darkCanvas = Color(0xFF0A0B0E);
  static const Color darkCard = Color(0xFF151821);
  static const Color darkCardElevated = Color(0xFF1E2230);
  static const Color darkBorder = Color(0xFF232838);

  // Editorial Pastel Palette (Light Mode)
  static const Color pastelCanvas = Color(0xFFF7F6F2);
  static const Color pastelCard = Color(0xFFFFFFFF);
  static const Color pastelBorder = Color(0xFFE2E8F0);
  static const Color inkBlack = Color(0xFF111827);
  static const Color inkBlackSoft = Color(0xFF374151);
  static const Color slateMuted = Color(0xFF64748B);
  static const Color slateLight = Color(0xFF94A3B8);

  // Semantic Pastel Accents
  static const Color pastelYellow = Color(0xFFFDE047);
  static const Color pastelLavender = Color(0xFFDDD6FE);
  static const Color pastelMint = Color(0xFFA7F3D0);
  static const Color pastelCoral = Color(0xFFFECDD3);

  // Financial Indicators
  static const Color income = Color(0xFF10B981);
  static const Color expense = Color(0xFFF43F5E);
  static const Color expenseSoft = Color(0x26F43F5E);
  static const Color textPrimary = Color(0xFFF8FAFC);
  static const Color textSecondary = Color(0xFF94A3B8);
}
```

5.2 Arched Navigation Dock & Shell (main_shell_screen.dart)

Implements the custom arched dock painter and symmetrical 4-tab shell with elevated '+' button:

Listing: lib/presentation/screens/main_shell_screen.dart (Excerpts)

```
class MainShellScreen extends StatefulWidget {
  const MainShellScreen({super.key});
  @override
  State<MainShellScreen> createState() => _MainShellScreenState();
}

class _MainShellScreenState extends State<MainShellScreen> {
```

```
  int _currentIndex = 0;
  final List<Widget> _screens = const [
    DashboardScreen(), StatsScreen(), TransactionsScreen(), ProfileScreen()
  ];

  @override
  Widget build(BuildContext context) {
    final isDark = Theme.of(context).brightness == Brightness.dark;
    final dockBg = isDark ? AppColors.darkCardElevated : Colors.white;
    final dockBorder = isDark ? AppColors.darkBorder : AppColors.pastelBorder;
    final plusBg = isDark ? AppColors.neonLime : AppColors.inkBlack;

    return Scaffold(
      backgroundColor: isDark ? AppColors.darkCanvas : AppColors.pastelCanvas,
      extendBody: false,
      body: IndexedStack(index: _currentIndex, children: _screens),
      bottomNavigationBar: SafeArea(
        top: false,
        child: Container(
          padding: const EdgeInsets.only(left: 18, right: 18, bottom: 12),
          color: isDark ? AppColors.darkCanvas : AppColors.pastelCanvas,
          child: SizedBox(
            height: 84,
            child: Stack(
              clipBehavior: Clip.none,
              children: [
                Positioned.fill(
                  child: CustomPaint(
                    painter: ArchedDockPainter(
                      backgroundColor: dockBg, borderColor: dockBorder,
                      cornerRadius: 28.0, domePeakHeight: 18.0, isDark: isDark,
                    ),
                  ),
                ),
                Positioned(
                  top: 18, left: 0, right: 0, bottom: 0,
                  child: Row(
                    children: [
                      Expanded(child: _buildNavItem(0, Icons.wallet_outlined, Icons.wallet, 'Home', isDark)),
                      Expanded(child: _buildNavItem(1, Icons.pie_chart_outline, Icons.pie_chart, 'Analytics',
isDark)),
                      const SizedBox(width: 68), // Spacer for dome
                      Expanded(child: _buildNavItem(2, Icons.receipt_long_outlined, Icons.receipt_long, 'History',
isDark)),
                      Expanded(child: _buildNavItem(3, Icons.person_outline, Icons.person, 'Profile', isDark)),
                    ],
                  ),
                ),
                Positioned(
                  top: 4, left: 0, right: 0,
                  child: Center(
                    child: GestureDetector(
                      onTap: () => Navigator.of(context).push(MaterialPageRoute(builder: (_) => const
AddEditExpenseScreen())),
                      child: Container(
                        width: 52, height: 52,
                        decoration: BoxDecoration(color: plusBg, shape: BoxShape.circle),
                        child: Icon(Icons.add_rounded, color: isDark ? AppColors.darkCanvas : Colors.white, size:
30),
                      ),
                    ),
                  ),
                ),
              ],
            ),
          ),
        ),
      ),
    );
  }
}
```

5.3 Daily Spending Trend Bar Chart (daily_spending_trend_chart.dart)

Calculates daily burn rate vs student allowance pace with spike detection and budget threshold lines:

Listing: lib/presentation/screens/stats/widgets/daily_spending_trend_chart.dart (Excerpts)

```
class DailySpendingTrendChart extends StatefulWidget {
  final List<Expense> expenses;
  final double monthlyBudget;
  final DateTime selectedMonth;
  const DailySpendingTrendChart({super.key, required this.expenses, required this.monthlyBudget, required
this.selectedMonth});
  @override
  State<DailySpendingTrendChart> createState() => _DailySpendingTrendChartState();
}

class _DailySpendingTrendChartState extends State<DailySpendingTrendChart> {
  bool _isWeeklyView = false;

  @override
  Widget build(BuildContext context) {
    final isDark = Theme.of(context).brightness == Brightness.dark;
    final dailyTarget = widget.monthlyBudget / 30.0;
    final cardBg = isDark ? AppColors.darkCard : Colors.white;
    final normalBarColor = isDark ? AppColors.neonLime : const Color(0xFF8B5CF6);
    final overBarColor = isDark ? AppColors.expense : const Color(0xFFF43F5E);

    return Container(
      padding: const EdgeInsets.all(20),
      decoration: BoxDecoration(
        color: cardBg,
        borderRadius: BorderRadius.circular(24),
        border: Border.all(color: isDark ? AppColors.darkBorder : AppColors.pastelBorder),
      ),
      child: Column(
        children: [
          Row(
            mainAxisAlignment: MainAxisAlignment.spaceBetween,
            children: [
              Text('DAILY OUTCOME BY WEEKDAY', style: TextStyle(fontSize: 11, fontWeight: FontWeight.w800)),
              Container(
                child: Row(
                  children: [
                    _buildTogglePill('Daily', !_isWeeklyView, isDark, () => setState(() => _isWeeklyView = false)),
                    _buildTogglePill('Weekly', _isWeeklyView, isDark, () => setState(() => _isWeeklyView = true)),
                  ],
                ),
              ),
            ],
          ),
          SizedBox(
            height: 180,
            child: BarChart(
              BarChartData(
                maxY: 6000,
                barGroups: List.generate(7, (i) => BarChartGroupData(
                  x: i,
                  barRods: [BarChartRodData(toY: weekdayTotals[i+1] ?? 0.0, color: (weekdayTotals[i+1] ?? 0.0) >
dailyTarget ? overBarColor : normalBarColor, width: 18)],
                )),
              ),
            ),
          ),
        ],
      ),
    );
  }
}
```

5.4 Tactile Dot Matrix Circular Chart (dot_matrix_pie_chart.dart)

Renders 44 discrete circular dots around a 240x240 circle representing category expenditure:

Listing: lib/presentation/screens/stats/widgets/dot_matrix_pie_chart.dart (Canvas Painter)

```
class _DotMatrixCanvasPainter extends CustomPainter {
  final List<CategorySpending> spendingList;
  final double totalSpent;
  final double budget;
  final int totalDots;
  final bool isDark;

  _DotMatrixCanvasPainter({required this.spendingList, required this.totalSpent, required this.budget, this.totalDots
= 44, required this.isDark});

  @override
  void paint(Canvas canvas, Size size) {
    final center = Offset(size.width / 2, size.height / 2);
    final radius = (size.width / 2) - 16;
    final dotPaint = Paint()..style = PaintingStyle.fill;
    final inactiveColor = isDark ? const Color(0xFF232838) : const Color(0xFFE2E8F0);

    for (int i = 0; i < totalDots; i++) {
      final angle = (2 * math.pi * (i / totalDots)) - (math.pi / 2);
      final dotCenter = Offset(center.dx + radius * math.cos(angle), center.dy + radius * math.sin(angle));
      final color = _getColorForDot(i) ?? inactiveColor;
      dotPaint.color = color;
      canvas.drawCircle(dotCenter, 4.5, dotPaint);
    }
  }

  Color? _getColorForDot(int index) {
    double runningFraction = 0.0;
    final dotFraction = index / totalDots;
    for (final cat in spendingList) {
      final frac = cat.percentage / 100.0;
      if (dotFraction >= runningFraction && dotFraction < (runningFraction + frac)) {
        return cat.category.color;
      }
      runningFraction += frac;
    }
    return null;
  }
  @override
  bool shouldRepaint(covariant CustomPainter oldDelegate) => true;
}
```

5.5 Handcrafted Vector Student Illustration (CustomPainter)

Vector custom painter producing the unclipped student study desk illustration on the splash screen:

Listing: lib/presentation/widgets/illustrations/editorial_student_splash_illustration.dart (Excerpts)

```
class _EditorialSplashPainter extends CustomPainter {
  @override
  void paint(Canvas canvas, Size size) {
    final w = size.width;
    final h = size.height;
    final inkPaint = Paint()..color = AppColors.inkBlack..style = PaintingStyle.stroke..strokeWidth = 2.4;

    // 1. Decorative lavender backdrop
    final bgRect = RRect.fromRectAndRadius(Rect.fromCenter(center: Offset(w * 0.5, h * 0.52), width: w * 0.9, height:
h * 0.85), const Radius.circular(32));
```

```
    canvas.drawRRect(bgRect, Paint()..color = AppColors.pastelLavender);
    canvas.drawRRect(bgRect, inkPaint);

    // 2. Desk surface & stacked textbooks
    final deskY = h * 0.78;
    canvas.drawLine(Offset(w * 0.1, deskY), Offset(w * 0.9, deskY), inkPaint..strokeWidth = 2.5);
    canvas.drawRRect(RRect.fromRectAndRadius(Rect.fromLTWH(w * 0.16, deskY - 14, 52, 14), const Radius.circular(3)),
Paint()..color = AppColors.pastelMint);
    canvas.drawRRect(RRect.fromRectAndRadius(Rect.fromLTWH(w * 0.18, deskY - 26, 46, 12), const Radius.circular(3)),
Paint()..color = AppColors.pastelYellow);

    // 3. Student torso with striped sweater
    final torsoPath = Path()..moveTo(w * 0.36, deskY)..lineTo(w * 0.38, h * 0.44)..quadraticBezierTo(w * 0.5, h *
0.42, w * 0.62, h * 0.44)..lineTo(w * 0.64, deskY)..close();
    canvas.drawPath(torsoPath, Paint()..color = AppColors.pastelYellow);
    canvas.drawPath(torsoPath, inkPaint);
    for (double sy = h * 0.48; sy < deskY - 6; sy += 9) {
      canvas.drawLine(Offset(w * 0.375, sy), Offset(w * 0.625, sy), inkPaint..strokeWidth = 1.8);
    }

    // 4. Glasses and hair bun
    canvas.drawCircle(Offset(w * 0.46, h * 0.32), 7, inkPaint);
    canvas.drawCircle(Offset(w * 0.54, h * 0.32), 7, inkPaint);
    canvas.drawCircle(Offset(w * 0.50, h * 0.17), 10, Paint()..color = AppColors.inkBlack);
  }
  @override
  bool shouldRepaint(covariant CustomPainter oldDelegate) => false;
}
```

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